- Data Structures in Python
    - List
    - Tuples
    - Dictionaries
    - Sets
        - Set Operations
- List Comprehension
    - Nested List comprehension
- Strings
    - Some special string functions
    - converting string to list
    - Flattening the List
- Python memory


```
a= 10
while a > 0:
  print(a)
  a = a-1
```

a → 10 9

while a > 0 :
    print(a)
    a = a - 1

10 > 0
↳ 10
a > 0 9

10
9
8
7
6
5
4
3
2
1

① Range

↳ range (a, b, c)

a = starting Point

b = ending Point

c = Step function

Include

Exclude

⇒ range (1, 6)

↳ 1, 2, 3, 4, 5

5, 6, 7, 8, 9, 10

1
2
3
4

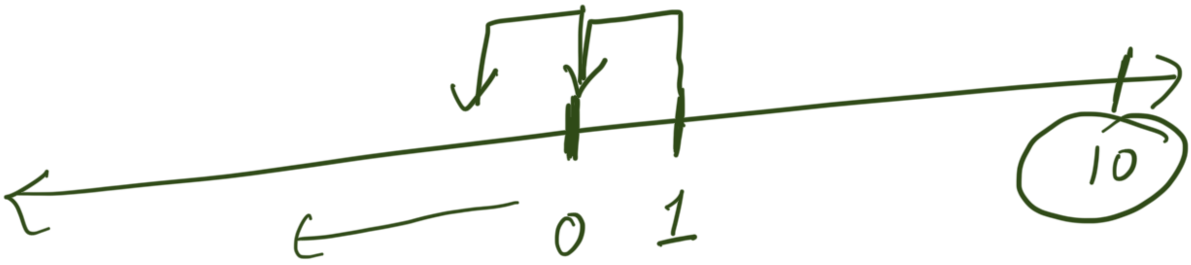
r (1, 10, 3)

⇒ range (...

⇒ range(5.5) ×

⇒ ?

5
6
7
8
9



range(10, 1, -1)

range(1, 10, -1)

f(x) = y

x = input

y = output

= function

98 °F → °C

↳ Piece of code that can be used multiple times

↳ it takes input and does something

↳ " on it.

def abc(a, b):

Data structures ↳ strings
↳ List
↳ Tuples
↳ Sets
↳ Dictionaries

① List : It's heterogenous collection of objects

a = ["amit", 37, 6.5, True, "Python"]
0 1 2 3 4

a[4] = Python

a[5] = ?

-1

-5	-4	-3	-2	
amit	10	17.5	False	python
0	1	2	3	4

a[4] = python

a[-1] = python

a[2:4]

a[2:]

↳ start = 2

end = len(a)

a[x:y:z]

x → start

y → end

z → jump

```
my_list = ["amit",10,19.2,True,"Python"]
my_list[-3:2:-1]
```

-5	-4	-3	-2	-1
amit	10	19.2	True	Python
0	1	2	3	4

Start -3 ⇒ 19.2

Stop → 2 ⇒ stop before index 2

Step -1 ⇒ go backwards

a =

-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
1	2	3	4	5	6	7	8	9	10
0	1	2	3	4	5	6	7	8	9

-3

a[-3 : 2 : -1]

② Tuples :-

Simply tuples are immutable lists

(   )

Packing and Unpacking in tuples

$$a = (1, 2, 3)$$

$$x, y, z = a$$

③ Dictionaries :-

↳ have key value pairs

↳ unordered

↳ mutable

↳ Indexable using keys

↳ Dictionaries are iterable

K, V

↳ Any immutable DS can be key

↳ Any immutable or mutable DS can be value

$$d = \{ K : V \}$$

Sets :- unordered

non-indexable

Mutable

* Stores only unique values

iterable

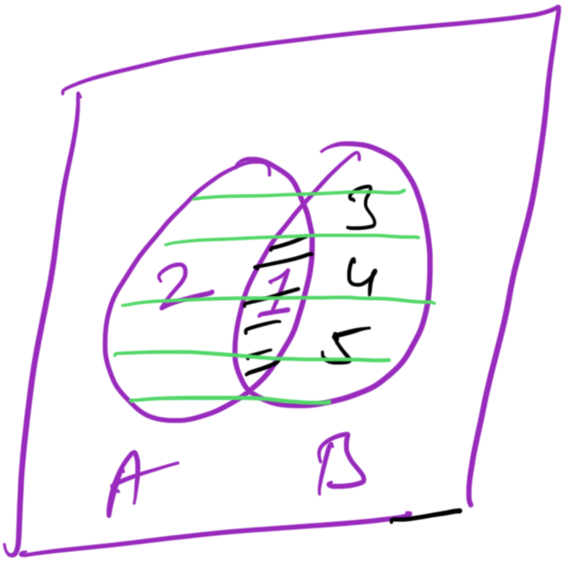
Can only contain immutable DS

{

{

A = 1, 2

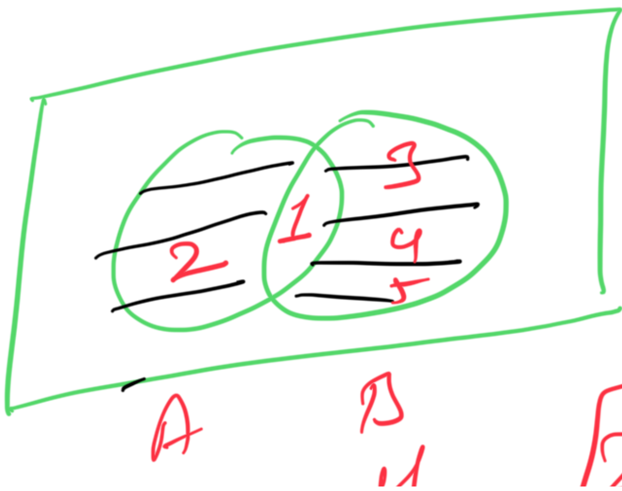
B = 1, 3, 4, 5



A. intersection(B) = 1

A. Union(B) = {1, 2, 3, 4, 5 }

A. Difference(B) = { 2 }



2, 3, 4, 5

Symmetric Diff =